Lab Report: Image classification experiments

1 Introduction

In this report we compare several models on standard benchmarks. Each experiment was repeated 3 times with different random seeds. We trained each model for 30 epochs with a batch size of 128.

2 Results

Figure 1 shows the test accuracy by dataset.

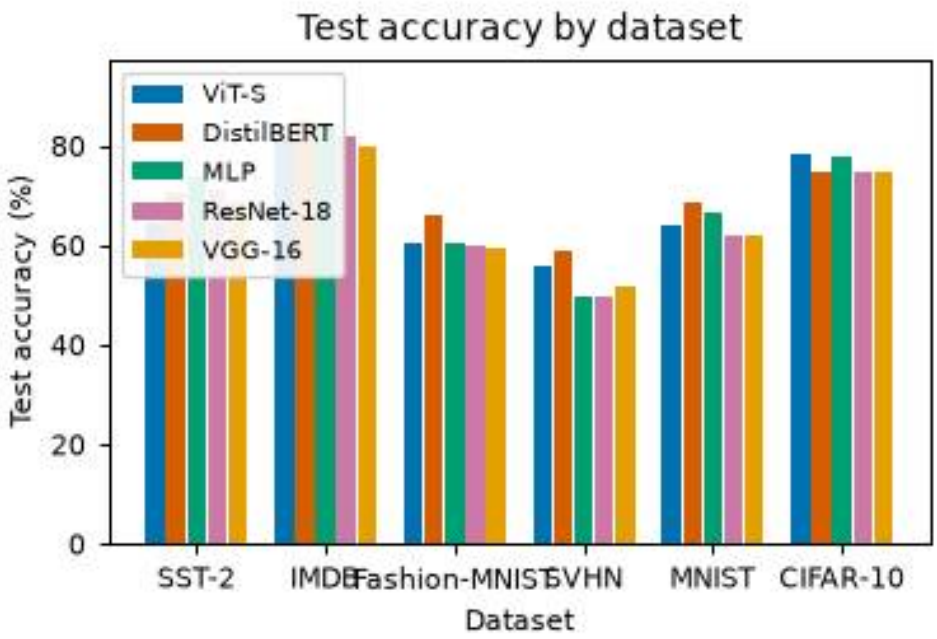


Figure 1: Test accuracy by dataset for the models in this experiment.

As shown in Figure 1, DistilBERT reaches 57.1% test accuracy on SVHN. On IMDB, MLP is 53.2 points better than VGG-16 (Figure 1). On Fashion-MNIST, MLP obtains a test accuracy of 60%. DistilBERT outperforms ViT-S by 3 percentage points on IMDB.

Figure 2 shows the accuracy during training.

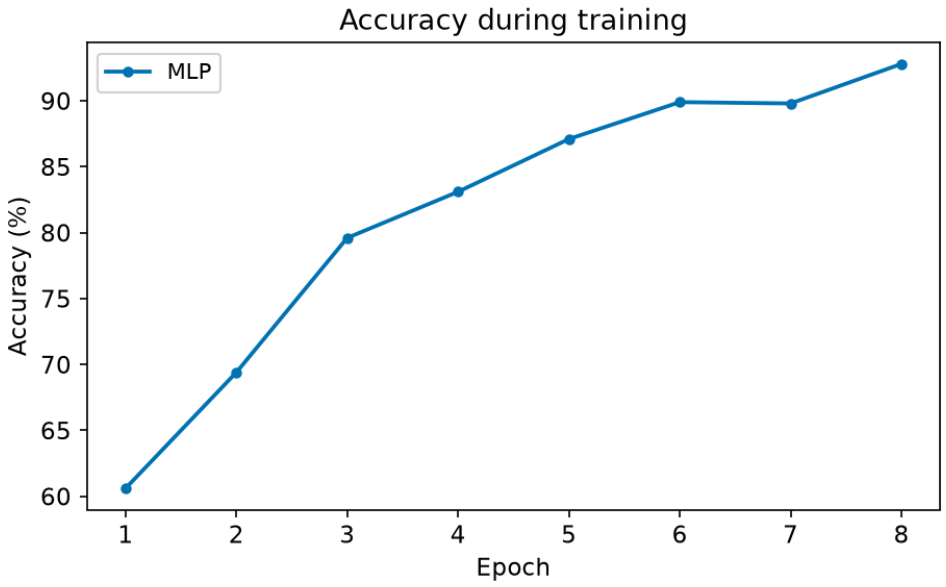


Figure 2: Accuracy during training for the models in this experiment.

MLP attains a accuracy of 99.9% at epoch 8. As shown in Figure 2, MLP reaches 83.1% accuracy at epoch

4. MLP attains a accuracy of 86.7% at epoch 3 (Figure 2). From epoch 5 to epoch 6, the accuracy of MLP improves by a relative 1.2%.

Figure 3 shows the training loss.

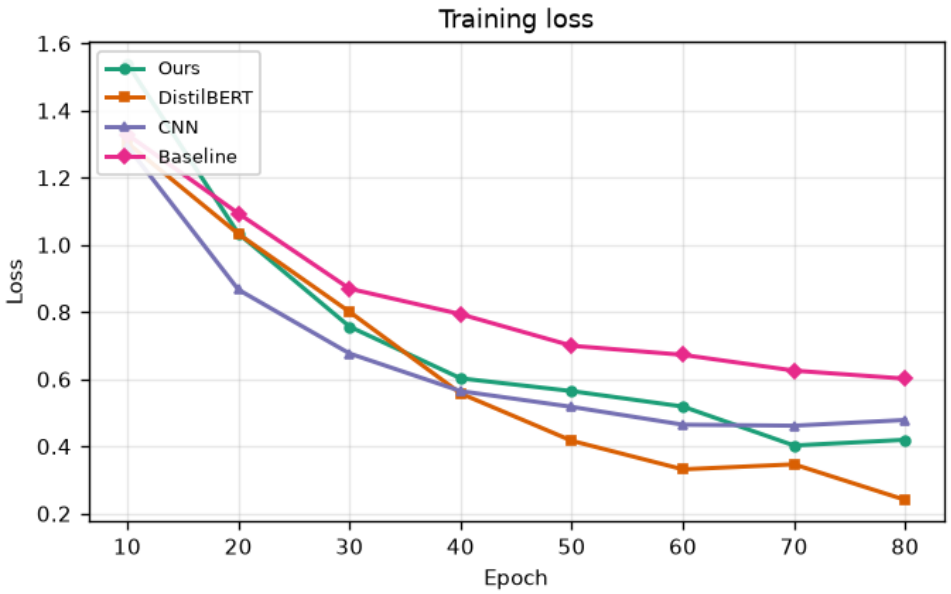


Figure 3: Training loss for the models in this experiment.

As shown in Figure 3, at epoch 50, the loss of Ours is 0.422. Baseline has a loss of 0.156 at epoch 30.

3 Discussion

XGBoost reaches 78.1% accuracy on Fashion-MNIST. We note that the training was stable for all models.